

CO1 CRYPTOGRAPHY

1, encrypt "Meet at NOON" using a shift of 4

(a) Shift each letter 4 positions forward

Plaintext	M	E	E	T	A	T	N	O	O	N
Ciphertext	Q	I	X	E	X	R	S	S	R	A

Encrypted message:-

Q I X E X R S S R A

(b) The authorized member knows that the message was encrypted using a shift of 4. To decrypt it, they shift each letter 4 positions backward word in the alphabet.

for example:-

Q → M

I → E

X → T

E → A

X → T

R → N

S → O

S → O

R → N

This restores the original message:-

MEET AT NOON

1) determine the original message

The ciphertext:-

WEIRL TEERDSO EEFEAOCAVDEN

is a well-known example of a rail fence cipher (3 rails)

by reversing the 3-rail transposition, the original plaintext is:-

WE ARE DISCOVERED FLEE AT ONCE

(b) why is this method vulnerable in modern communication systems?

The rail fence cipher is considered insecure because:-

- It does not change the letters, only their positions
- An attacker can easily try different numbers of rails (brute force).
- It is vulnerable to frequency analysis and pattern recognition
- Modern Computers can decrypt such ciphers within seconds.
- It provides very weak security for confidential communication

(c) one practical ~~extra~~ enhancement:-
Practical improvement is to replace the rail fence cipher with a modern encryption algorithm such as AES (Advanced Encryption standard).

Benefits of AES:-

- uses strong cryptographic keys (128, 192, or 256 bits)
- Extremely difficult to crack using current technology
- widely used for secure communication in banking, government, and online services.
- provides much stronger confidentiality than classical transposition ciphers